

What Is Streamlit & Why Should Students learn it?

Streamlit is an *open-source Python library* that lets you build interactive web apps for data science and machine learning using just Python—no need for HTML, CSS, or JavaScript. ***You can quickly prototype applications and expose in a web browser, for example.***

Step-by-Step Beginner Tutorial

1. Install Streamlit

Make sure Python and pip are installed. Then run:

```
pip install streamlit
```

2. Create Your First App

Create a file called [app.py](#) and add:

```
import streamlit as st
```

```
st.title("Hello Streamlit!")
```

```
st.write("Welcome to your first Streamlit app.")
```

3. Run the App

In your terminal or command prompt:

```
streamlit run app.py
```

This will launch a local server and open the app in your browser.

4. Add Interactivity

Try adding widgets:

```
name = st.text_input("Enter your name:")
```

```
st.write(f"Hello, {name}!")
```

5. Display Data

You can show dataframes and charts easily:

```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.DataFrame(np.random.randn(10, 2), columns=["A", "B"])
st.line_chart(df)
```

6. Create Multipage Apps

Use [st.sidebar](#) for navigation and organize your app into multiple files.

7. A sample Python source code

```
import streamlit as st
```

```
st.title("Hello CSE3rd Year Students!")
st.write("Refer to this source file to get started with Streamlit.")
```

```
st.write("This is text input field")
name = st.text_input("Enter your name")
if name:
    st.write(f"Hello, {name}!")
```

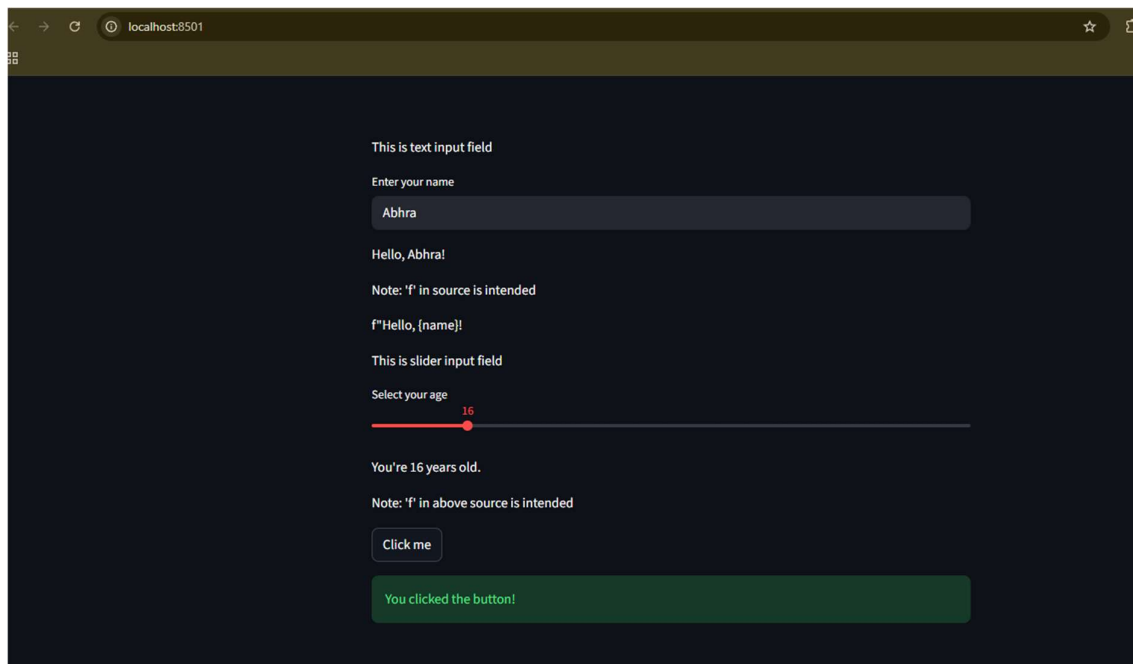
```
st.write("""Note: 'f' in source is intended
```

```
f"Hello, {name}!"
```

```
""") # This is a multiline string
```

```
st.write("This is slider input field")
age = st.slider("Select your age", 0, 100, 21)
st.write(f"You're {age} years old.")
st.write("Note: 'f' in above source is intended")
```

```
if st.button("Click me"):
    st.success("You clicked the button!")
```

Output:

More Learning Resources

- [Streamlit Docs – Tutorials](#) cover authentication, theming, data sources, and multipage apps¹.
- [GeeksforGeeks Beginner's Guide](#) explains installation, basic UI, and running apps².
- [Dev-Kit's UI Tutorial](#) walks through building a simple user interface step-by-step³.

References (3)

¹*Tutorials - Streamlit Docs.* <https://docs.streamlit.io/develop/tutorials>

²*A Beginners Guide To Streamlit - GeeksforGeeks.*
<https://www.geeksforgeeks.org/python/a-beginners-guide-to-streamlit/>

³*Streamlit Beginner's Guide to Creating a Basic UI: A Step-by-Step Tutorial.* <https://dev-kit.io/blog/python/streamlit-ui-development>